

Ritesh Somashekar

Python/UI Developer

Fairfax, VA | (571) 245-7770 | ritesh.chandra33@gmail.com | LinkedIn

PROFESSIONAL SUMMARY

- **Python/UI Developer** with **6+ years** of experience enhancing and modernizing enterprise platforms, building **modern, scalable web user interfaces, distributed data-processing pipelines, and containerized applications** using **Python, Java, React, and FastAPI**. Proven expertise in **access control and authorization (OAuth 2.0, RBAC), job queuing and scheduling, and workflow orchestration (Apache Airflow)**.
- Strong background in **profiling and optimizing long-running compute workloads** with **Apache Spark and PySpark**, defining **performance metrics and statistics**, and refactoring orchestration logic to improve **execution efficiency and throughput**. Experienced in **Docker, Kubernetes, and CI/CD pipelines** on **AWS and Azure**, delivering secure, **HIPAA-compliant** systems.
- Demonstrated success in **redesigning UI architecture** to improve **information hierarchy, usability, consistency, and layout** across functional modules. Adept at **technical design discussions, code reviews, and technical documentation**, collaborating effectively across engineering, product, and research teams.

TECHNICAL SKILLS

Programming & Scripting: Python (pandas, NumPy, asyncio), Java, JavaScript, TypeScript, SQL (PostgreSQL, Oracle), Bash

Web UI & Frontend: React, HTML5, CSS3, Responsive Design, Component Libraries, Design Systems, Figma, Accessibility (WCAG), Data Visualization (Plotly, Power BI, Tableau)

Backend, APIs & Security: FastAPI, Flask, RESTful Microservices, API Versioning, OAuth 2.0, JWT, Role-Based Access Control (RBAC), Authentication & Authorization

Job Scheduling & Orchestration: Apache Airflow (DAGs), Celery, Job Queuing & Prioritization, Concurrent Execution, Kafka, Redis, Event-Driven Workflows

Containerization & DevOps: Docker, Kubernetes, Jenkins, CI/CD, Git, DVC, MLflow

Distributed Processing & Performance: Apache Spark, PySpark, Hadoop, Hive, ETL Pipelines, Profiling (cProfile), Performance Metrics, Throughput Optimization

Cloud & Databases: AWS (S3, EC2, Lambda, Glue, RDS, SageMaker), Azure (Stream Analytics, Data Services), MongoDB, Snowflake, Delta Lake

ML & Analytics: PyTorch, scikit-learn, HuggingFace Transformers, BERT/BioBERT, Optuna

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA.

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India.

PROFESSIONAL EXPERIENCE

George Mason University | UI/UX Developer

Fairfax, VA | August 2024 – Present

- Modernized the UI architecture of a **multi-user research analytics dashboard** using **React** and **Python (FastAPI)**, redesigning **information hierarchy, navigation, and layout** across functional modules to improve usability and consistency.
- Built a reusable **component library and design system** (typography, spacing, interaction states) applying **responsive design** and **accessibility (WCAG)** practices, standardizing UI patterns across all dashboard views.
- Implemented **role-based access control (RBAC)** and token-based **authorization** on **FastAPI** endpoints, securing multi-user access to datasets, models, and results by role and project.
- Designed a **job queuing and scheduling** layer for long-running processing jobs over **30K+ documents**, supporting **concurrent execution, prioritization**, and real-time job-status tracking surfaced in the UI.
- Profiled and optimized end-to-end **run-processing pipelines** using **Python, PyTorch, MLflow, and Optuna**, cutting iteration cycles from **3 days to under 18 hours** and serving predictions to the UI with **sub-200ms latency**.
- Streamlined data-preparation workflows with **Pandas** and **Scikit-learn**, standardizing **15GB** of structured and unstructured data across 6 datasets and reducing preparation time by **30%**.
- Containerized services with **Docker** and automated releases through **CI/CD** pipelines, reducing release cycles from **2 weeks to 5 days** and enabling **A/B testing** of UI and model variants.

- Led **design and code reviews** using **Git** and **DVC**, and authored **technical documentation** and engineering best practices, reducing onboarding time for new contributors to **under 2 days**.

Carelon Global Solutions | Python Developer (Client: Accenture)

India | January 2019 – July 2023

- Engineered **access control and authorization** for enterprise **RESTful microservices** using **OAuth 2.0**, role-based permissions, and **API versioning**, securing **HIPAA-compliant** healthcare data exchange for a U.S. healthcare client.
- Orchestrated end-to-end workflows with **Apache Airflow DAGs** for data ingestion, transformation, and model retraining; refactored **scheduling and dependency logic** to support **concurrent execution and prioritization**, reducing manual effort by **25%**.
- Profiled and optimized large-scale **distributed processing workloads** by migrating from **Hive** to **Apache Spark/PySpark**, tuning partitioning and job execution to significantly reduce processing time and increase throughput.
- Built scalable **ETL pipelines** using **PySpark** and **AWS (S3, Glue, SageMaker)** with **Delta Lake**, improving data reliability by **30%** across ingestion and processing stages.
- Containerized and deployed Python services using **Docker, Kubernetes, and Jenkins CI/CD**, improving system uptime by **20%** and enabling real-time EHR integration via **REST APIs**.
- Developed real-time monitoring pipelines and **interactive Power BI dashboards** on **Azure Stream Analytics**, surfacing **performance metrics and statistics** that reduced unplanned downtime by **20%**.
- Delivered Python NLP services using **BERT** and **BioBERT** for clinical entity extraction, achieving **80% accuracy** and reducing document processing time by **55%**.
- Designed **MongoDB** data models and **Python/Java** validation scripts (UDFs) for high-throughput, fault-tolerant pipelines, improving data quality and consistency.
- Collaborated with Data Engineering, Product, and **Accenture** client stakeholders in **technical design discussions and code reviews** to define use cases, KPIs, and delivery roadmaps, reducing project turnaround time by **15%**.